

Course 50.050/50.550

Advanced Algorithms

Week 4 – Lecture L04.02



Outline of Lecture

- ▶ Subgraphs, induced subgraphs
- ▶ Graph colorings, chromatic number, edge chromatic number
- ▶ Brooks' theorem, Vizing's theorem, four color theorem
- ▶ Properties of bipartite graphs

Subgraphs

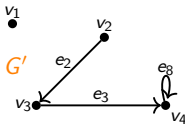
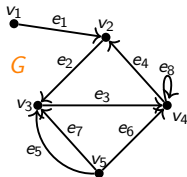
Recall: A graph is a 3-tuple $G = (V, E, \phi)$, where V is the vertex set, E is the edge set, and ϕ is the incidence function, with domain E .

- ▶ We can consider subsets of sets, so analogously, we can also consider subgraphs of graphs.

Definition: Let $G = (V, E, \phi)$ and $G' = (V', E', \phi')$ be graphs. If $V' \subseteq V$, $E' \subseteq E$, and if $\phi'(e) = \phi(e)$ for every edge $e \in E'$, then we say that G' is a **subgraph** of G .

- ▶ **Intuition:** If G' is a subgraph of G , then every edge of G' must also be an edge of G .
- ▶ We can think of obtaining G' from G by deleting vertices and edges from G , with the condition that if e is an undeleted edge (i.e. e is an edge of G'), then the endpoints of e cannot be deleted.

Example: G' is a subgraph of G .



$$V(G) = \{v_1, v_2, v_3, v_4, v_5\}.$$

$$E(G) = \{e_1, e_2, \dots, e_8\}.$$

$$V(G') = \{v_1, v_2, v_3, v_4\}.$$

$$E(G') = \{e_2, e_3, e_8\}.$$

Induced subgraphs

Definition: Let $G = (V, E, \phi)$ be a graph, and let $U \subseteq V$ be a subset.

- ▶ Among all possible subgraphs with a fixed vertex set U , there is a unique subgraph with the **largest possible edge set**. We call this subgraph the **vertex-induced subgraph** of G **induced by** U .
 - ▶ **Notation:** This induced subgraph is usually written as $G[U]$.
 - ▶ A **vertex-induced subgraph** of G is a subgraph G' of G such that $G' = G[U]$ for some subset $U \subseteq V$.

Intuition: $G[U]$ is a subgraph of G with vertex set U , such that all edges of G whose endpoints are in U are also edges of $G[U]$.

Definition: Let $G = (V, E, \phi)$ be a graph, and let $D \subseteq E$ be a subset.

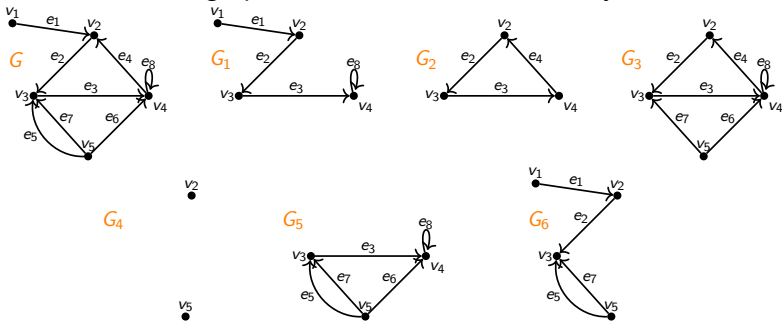
- ▶ Among all possible subgraphs with a fixed edge set D , there is a unique subgraph with the **smallest possible vertex set**. We call this subgraph the **edge-induced subgraph** of G **induced by** D .
 - ▶ **Notation:** This induced subgraph is sometimes written as $G[D]$.
 - ▶ An **edge-induced subgraph** of G is a subgraph G' of G such that $G' = G[D]$ for some subset $D \subseteq E$.

Intuition: $G[D]$ is a subgraph of G with edge set D , such that only vertices that are endpoints of edges in D become vertices of $G[D]$.



Examples of vertex-induced subgraphs

Example: Let $G = (V, E, \phi)$ be a graph as depicted below. There are six subgraphs of G shown below, namely $G_1, \dots, G_6$.

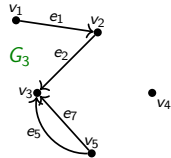
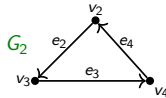
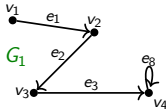
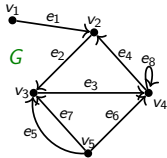


- ▶ G_1, G_2, G_3 are not vertex-induced subgraphs of G .
 - ▶ $e_4 \notin E(G_1)$, but e_4 has endpoints v_2 and v_4 , which are in $V(G_1)$.
 - ▶ $e_8 \notin E(G_2)$, but e_8 has endpoint v_4 , which is in $V(G_2)$.
 - ▶ $e_5 \notin E(G_3)$, but e_5 has endpoints v_3 and v_5 , which are in $V(G_3)$.
- ▶ G_4, G_5 and G_6 are vertex-induced subgraphs of G .
 - ▶ $G_4 = G[\{v_2, v_5\}]$, $G_5 = G[\{v_3, v_4, v_5\}]$, $G_6 = G[\{v_1, v_2, v_3, v_5\}]$.



Examples of edge-induced subgraphs

Example: Let $G = (V, E, \phi)$ be a graph as depicted below. There are three subgraphs of G shown below, namely G_1 , G_2 , G_3 .



- ▶ G_3 is not an edge-induced subgraph of G .
 - ▶ Deleting the isolated vertex v_4 from G_3 still yields a subgraph of G with the same edge set as $E(G_3)$.
 - ▶ A vertex in a graph is called an **isolated vertex** if it is not incident to any edges in the graph.
- ▶ G_1 and G_2 are edge-induced subgraphs of G .
 - ▶ $G_1 = G[\{e_1, e_2, e_3, e_8\}]$, $G_2 = G[\{e_2, e_3, e_4\}]$.

Fact: An edge-induced subgraph of G is precisely a subgraph of G with no isolated vertices.

Note: Because edge-induced subgraphs are “less interesting”, many authors write “induced subgraph” (without specifying whether it is vertex-induced or edge-induced) to mean “vertex-induced subgraph”.

Graph colorings

Idea: For graphs, we can color their vertices, and color their edges.

- ▶ Also, for plane graphs, we can color their faces.

Definition: Let $G = (V, E, \phi)$ be an **undirected** graph with **no loops**. Let C be a fixed non-empty set, e.g. $C = \{\text{'red'}, \text{'blue'}, \text{'green'}\}$, which we shall call the **color set**, and whose elements we shall call “**colors**”.

- ▶ A **vertex coloring** of G (w.r.t. C) is a function $\varphi : V \rightarrow C$ such that for every $e \in E$, the pair $\phi(e) = \{u, v\}$ satisfies $\varphi(u) \neq \varphi(v)$.
- ▶ An **edge coloring** of G (w.r.t. C) is a function $\varphi : E \rightarrow C$ such that for every two distinct edges $e, e' \in E$ incident to at least one common vertex (i.e. $|\phi(e) \cap \phi(e')| \neq 0$), we have $\varphi(e) \neq \varphi(e')$.

Intuition:

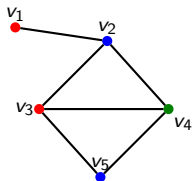
- ▶ A vertex coloring is an **assignment of colors to the vertices**, such that **adjacent vertices** are assigned **different colors**.
- ▶ An edge coloring is an **assignment of colors to the edges**, such that any two edges incident to a common vertex are assigned **different colors**.

- ▶ **Note:** Some authors use “incident edges” to mean edges that are incident to a common vertex, so you may see in some sources the definition of edge colorings in terms of “incident edges”.

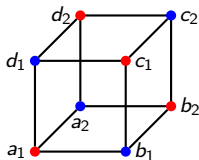


Examples of graph colorings

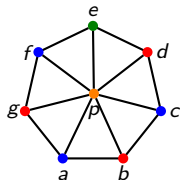
Examples of vertex colorings:



(3 colors used)

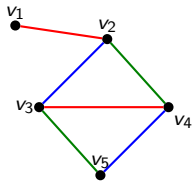


(2 colors used)

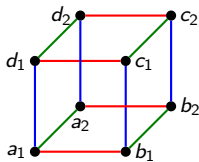


(4 colors used)

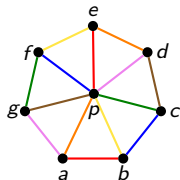
Examples of edge colorings:



(3 colors used)



(3 colors used)



(7 colors used)

More vertex coloring terminology

Important Note: There is nothing inherently special about the color set C . We could even let positive integers $1, 2, 3, \dots$ be the “colors”.

- ▶ Usually, we are interested only in the number of colors used, not the specific colors used.

Definition: Let $G = (V, E, \phi)$ be an **undirected** graph with **no loops**.

- ▶ Suppose $\varphi : V \rightarrow C$ is a vertex coloring of G for some set C . If among the colors in $\{\varphi(v) | v \in V\}$, there are exactly k distinct colors, then we say that φ is a **vertex k -coloring**.
 - ▶ **Note:** Many authors simply say that φ is a **k -coloring**.
 - ▶ **Note:** Some authors use “proper” to emphasize that adjacent vertices have different colors, i.e. “proper vertex k -coloring” or “proper k -coloring”.
- ▶ We say that G is **vertex k -colorable**, or simply **k -colorable**, if there exists a vertex k -coloring for G .
- ▶ We say that G is **vertex k -chromatic**, or simply **k -chromatic**, if G is vertex k -colorable, and not vertex k' -colorable for any $k' < k$.
- ▶ If G is k -chromatic, then we say that the **chromatic number** of G , denoted by $\chi(G)$, equals k .
 - ▶ χ is the Greek letter “chi”, and pronounced as “kai”.

Useful Note: The chromatic number is a graph invariant.



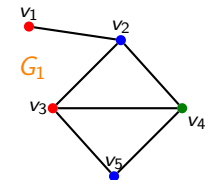
More on chromatic numbers

Intuition: The **chromatic number** of an undirected graph G with no loops is the **least number of colors** needed for a vertex coloring of G .

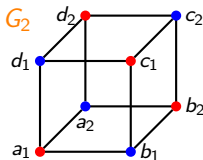
Theorem: For all $n \in \mathbb{Z}^+$, the complete graph K_n has chromatic number n . (Can you see why this is true?)

Corollary: Let G be an undirected graph with no loops. If G contains K_n as a subgraph for some $n \in \mathbb{Z}^+$, then $\chi(G) \geq n$.

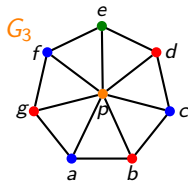
Example: Consider the following graphs G_1 , G_2 , G_3 .



(vertex 3-coloring)



(vertex 2-coloring)



(vertex 4-coloring)

- ▶ By construction, we have $\chi(G_1) \leq 3$, $\chi(G_2) \leq 2$, $\chi(G_3) \leq 4$.
- ▶ Since G_1, G_2, G_3 contain K_3, K_2, K_3 as subgraphs, respectively, we have $\chi(G_1) \geq 3$, $\chi(G_2) \geq 2$, $\chi(G_3) \geq 3$.

Question: Can you see why $\chi(G_3) \neq 3$? (Hint: Proof by contradiction.)



More edge coloring terminology

Definition: Let $G = (V, E)$ be an **undirected** graph with **no loops**.

- ▶ Suppose $\varphi : E \rightarrow C$ is an edge coloring of G for some set C .
If among the colors in $\{\varphi(e) | e \in E\}$, there are exactly k distinct colors, then we say that φ is an **edge k -coloring**.
 - ▶ **Note:** Some authors use “proper” to emphasize that edges sharing a common endpoint have different colors, i.e. “proper edge k -coloring”.
- ▶ G is called **edge k -colorable** if there is an edge k -coloring for G .
- ▶ G is called **edge k -chromatic** if G is edge k -colorable, and not edge k' -colorable for any $k' < k$.
- ▶ If G is edge k -chromatic, then we say that the **edge chromatic number** (or **chromatic index**) of G , denoted by $\chi'(G)$, equals k .
 - ▶ $\chi'(G)$ is the smallest number of colors needed for an edge coloring.

Notation: Given an undirected graph G , the maximum degree of the vertices of G is commonly denoted by $\Delta(G)$.

- ▶ Δ is the capital Greek letter “Delta”.

Useful Note: The edge chromatic number is a graph invariant.

Fact: If G is an undirected graph with no loops, then $\chi'(G) \geq \Delta(G)$.

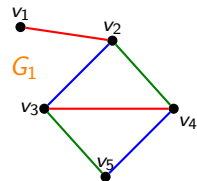
- ▶ Can you see why this is true? (*Hint: Proof by contradiction.*)



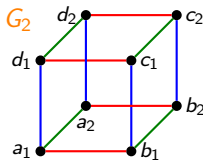
More on edge chromatic numbers

Recall: The **edge chromatic number** $\chi'(G)$ of a simple undirected graph G , which by definition is the **least number of colors** needed for an edge coloring of G , must be at least the maximum degree $\Delta(G)$.

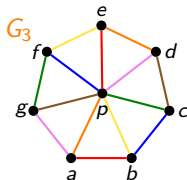
Example: Consider the following graphs G_1 , G_2 , G_3 .



(edge 3-coloring)



(edge 3-coloring)



(edge 7-coloring)

- ▶ By construction, we have $\chi'(G_1) \leq 3$, $\chi'(G_2) \leq 3$, $\chi'(G_3) \leq 7$.
- ▶ Since the maximum degrees are $\Delta(G_1) = 3$, $\Delta(G_2) = 3$, $\Delta(G_3) = 7$, we have $\chi'(G_1) \geq 3$, $\chi'(G_2) \geq 3$, $\chi'(G_3) \geq 7$.
- ▶ Therefore, $\chi'(G_1) = 3$, $\chi'(G_2) = 3$, $\chi'(G_3) = 7$.

Vizing's theorem

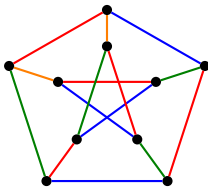
Theorem: (**Vizing's theorem** (1964)) If G is a **simple undirected** graph, then $\chi'(G) \leq \Delta(G) + 1$.

- ▶ Vizing's theorem can be proven by induction.

Corollary: Every simple undirected graph G satisfies

$$\Delta(G) \leq \chi'(G) \leq \Delta(G) + 1.$$

Example: An edge 4-coloring of the **Petersen graph** is depicted below.



- ▶ Let G denote the Petersen graph (as depicted above).
 - ▶ Note that $\Delta(G) = 3$. In fact, all vertices of G have degree 3, so G is a 3-regular graph. Thus, $\chi'(G) \geq 3$.
- ▶ By Vizing's theorem, we then have $3 \leq \chi'(G) \leq 4$.
- ▶ **Challenge:** Can you see why the Petersen graph is not **edge 3-colorable**? (In contrast, this graph is vertex 3-colorable.)



Brooks' theorem

Question: Can we find a theorem similar to Vizing's theorem, but for chromatic numbers? Yes!

Theorem: (Brooks' theorem (1941)) Let G be a **simple undirected** graph. If G is not a complete graph, and G is not an n -cycle for odd n , then $\chi(G) \leq \Delta(G)$.

Note: Brooks' theorem excludes complete graphs and odd cycles, whose chromatic numbers are known:

▶ **Theorem:** For all $n \in \mathbb{Z}^+$, $\chi(K_n) = n$.

▶ **Note:** $\Delta(K_n) = n - 1$. Hence $\chi(K_n) = \Delta(K_n) + 1$.

▶ **Theorem:** For all integers $n \geq 3$, $\chi(C_n) = \begin{cases} 3, & \text{if } n \text{ is odd;} \\ 2, & \text{if } n \text{ is even.} \end{cases}$

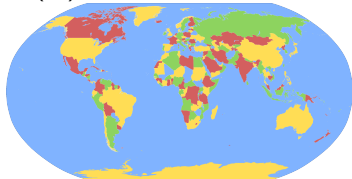
▶ **Note:** For $n \geq 3$, we have $\Delta(C_n) = 2$. (In fact, C_n is 2-regular.)

▶ Thus, if n is odd, then $\chi(C_n) = \Delta(C_n) + 1$.

Four color theorem

Theorem: (**Four color theorem** (1977)) Every undirected planar graph G with no loops is 4-colorable, i.e. $\chi(G) \leq 4$.

- ▶ **Note:** This is a theorem about vertex colorings of planar graphs.
- ▶ **Question:** What has this got to do with coloring regions on a world map?



Relating 4-colorability to coloring regions on a flat world map

- ▶ We can treat the map as a plane graph (G, Ψ) .
 - ▶ i.e. G is a planar graph, and Ψ is a plane drawing of G .
 - ▶ The regions on the map are the faces of the plane drawing Ψ .
- ▶ Consider dual graph G^* of this plane graph (G, Ψ) .
 - ▶ The dual graph of every plane graph is planar, so G^* is planar.
 - ▶ For any “proper” world map, the planar graph G^* has no loops.
 - ▶ *How can you get loops in dual graphs?*
- ▶ By the four color theorem, G^* is 4-colorable.
 - ▶ This means we can color the regions of the world map so that any two regions sharing a common boundary of non-zero length (i.e. depicting an edge of G) have different colors.



Recap: Bipartite graphs

Recall: (L03.02) A simple undirected graph $G = (V, E)$ is called a **bipartite** graph if there exists a partition of V into two subsets V_1, V_2 , such that every edge is of the form $\{v_1, v_2\}$ for some $v_1 \in V_1, v_2 \in V_2$.

► (V_1, V_2) is called a **bipartition** of our bipartite graph G .

Example: A bipartite graph with two bipartitions $(V_1, V_2), (V'_1, V'_2)$.



Extended Definition: A graph $G = (V, E, \phi)$ is **bipartite** if there exists two disjoint subsets V_1, V_2 of V whose union is $V_1 \cup V_2 = V$, such that for every edge $e \in E$, $\phi(e)$ contains one vertex in V_1 and contains one vertex in V_2 , i.e.

- (If G is undirected): $|\phi(e) \cap V_1| = |\phi(e) \cap V_2| = 1$.
- (If G is directed): Either $\phi(e) \in V_1 \times V_2$, or $\phi(e) \in V_2 \times V_1$.

Note: A bipartite graph cannot contain any loops.

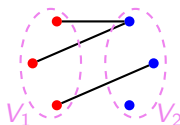
- **Note:** Many authors define “bipartite” for only simple graphs. Some authors insist “bipartite” are for simple undirected graphs only.



Bipartite graphs and colorings

Important Observation: If G is an undirected bipartite graph, then there exists a vertex 2-coloring for G .

- ▶ If (V_1, V_2) is a bipartition for G , then the map $\varphi : V(G) \rightarrow \{\text{'red'}, \text{'blue'}\}$ given by
$$\begin{cases} v \mapsto \text{'red'}, & \text{if } v \in V_1; \\ v \mapsto \text{'blue'}, & \text{if } v \in V_2; \end{cases}$$
 is a vertex 2-coloring for G .



- ▶ **Consequence:** Every undirected bipartite graph G is 2-colorable.
 - ▶ Every undirected bipartite graph G has chromatic number $\chi(G) \leq 2$.
 - ▶ If G is not a null graph, then $\chi(G) = 2$.

Theorem: Every 2-colorable undirected graph with no loops is bipartite.

Proof: Let G be a 2-colorable undirected graph, and let $\varphi : V(G) \rightarrow \{1, 2\}$ be a 2-coloring for G . Define the sets $V_i := \{v \in V \mid \varphi(v) = i\}$, for $i = 1, 2$. Then G is a bipartite graph with bipartition (V_1, V_2) . $\square$

For **undirected** graphs with **no loops**,
“**bipartite**” is synonymous with “**2-colorable**”.

Bipartite graphs and odd cycles

Recall: If $n \geq 3$ is an odd integer, then $\chi(C_n) = 3$.

- ▶ A cycle graph C_n with odd n is called an **odd cycle**.
 - ▶ **Recall:** C_1 has a single loop. 
- ▶ We know that a bipartite graph cannot have C_1 as a subgraph.
 - ▶ By definition, a bipartite graph cannot have any loops.

Theorem: Let $G = (V, E, \phi)$ be an undirected graph. Then G is bipartite if and only if no subgraph of G is an odd cycle.

*Before we prove this theorem, let's introduce another very useful idea in graph theory: the idea of **distance between vertices**.*

Definition: Let $G = (V, E, \phi)$ be an **undirected** graph. Let $u, v \in V$.

- ▶ Suppose there exists a sequence $(v_0, v_1, \dots, v_k)$ (for $k \in \mathbb{N}$) of vertices in V , such that $v_0 = u$ and $v_k = v$, and every two consecutive vertices v_{i-1}, v_i (for $1 \leq i \leq k$) are adjacent. Then among all such possible sequences, the smallest possible corresponding value of k is called the **distance** from u to v , which we denote by $d(u, v) = k$.
- ▶ If no such sequence exists, then we say that the **distance** from u to v is ∞ , which we denote by $d(u, v) = \infty$.



Bipartite graphs and odd cycles (continued)

Theorem: Let $G = (V, E, \phi)$ be an undirected graph. Then G is bipartite if and only if no subgraph of G is an odd cycle.

Proof: We split our proof into two parts.

- ▶ [Part 1]: If G is bipartite, then no subgraph of G is an odd cycle.
- ▶ [Part 2]: If no subgraph of G is an odd cycle, then G is bipartite.

[Part 1]: Assume G is bipartite. Then G is 2-colorable, with no loops.

- ▶ Let $\varphi : V(G) \rightarrow \{1, 2\}$ be a 2-coloring for G .

(Proof by contradiction) Suppose on the contrary there exists an odd cycle C_n (with $n \geq 3$) as a subgraph $G' = (V', E', \phi')$ of G .

- ▶ The restriction $\varphi|_{V'} : V' \rightarrow \{1, 2\}$ is a 2-coloring for G' .
 - ▶ (i.e. $v \mapsto \varphi(v)$ for all $v \in V'$)
 - ▶ This implies that $\chi(G') \leq 2$.
 - ▶ Yet $G' = C_n$ is an odd cycle, so $\chi(G') = 3$. Contradiction.

- ▶ Thus, G cannot have an odd cycle as a subgraph.
- ▶ This completes the proof for [Part 1].



Bipartite graphs and odd cycles (continued)

[Part 2]: Our goal is to show that the proposition “If no subgraph of G is an odd cycle, then G is bipartite” is true.

Key Idea 1: We shall assign colors to every vertex of G , either **red** or **blue**, via the following algorithm:

function COLORVERTICES(G)

Require: $G = (V, E, \phi)$ is an undirected graph.

- 1: **while** there exists an uncolored vertex in G **do**
 - 2: Select any uncolored vertex u in G .
 - 3: Color all vertices v with **even** finite distance $d(u, v)$ with color “**red**”.
 - 4: Color all vertices v with **odd** finite distance $d(u, v)$ with color “**blue**”.
- ▶ In each iteration of the while loop, all vertices reachable from the selected uncolored vertex u (including u itself) will be colored.
 - ▶ Since any finite $d(u, v)$ is either odd or even, not both, it is not possible for a red vertex to get re-colored blue.
 - ▶ After running COLORVERTICES(G), all vertices will be colored.

Key Idea 2: This color assignment is a vertex 2-coloring for G .

- ▶ If we can show this, then by definition, G is 2-colorable, and hence bipartite.

Bipartite graphs and odd cycles (continued)

Goal: Show that our color assignment is a vertex 2-coloring for G .

- ▶ **Subgoal 1:** Show that no two red vertices are adjacent.
- ▶ **Subgoal 2:** Show that no two blue vertices are adjacent.

Approach: (Proof by contradiction) Suppose on the contrary that there exist two adjacent vertices v, v' of the same color.

- ▶ Since v, v' are adjacent, they must have been colored in the same iteration of the while loop.
 - ▶ Let u be the selected vertex for this iteration.
 - ▶ Then both $d(u, v)$ and $d(u, v')$ must have the same parity.
 - ▶ There is a sequence $(v_0, v_1, \dots, v_k)$ of vertices in V such that $v_0 = u, v_k = v$, such that consecutive vertices are adjacent.
 - ▶ For $1 \leq i \leq k$, let e_i be an edge in E such that $\phi(e_i) = \{v_{i-1}, v_i\}$.
 - ▶ There is a sequence $(v'_0, v'_1, \dots, v'_\ell)$ of vertices in V such that $v'_0 = u, v'_\ell = v'$, such that consecutive vertices are adjacent.
 - ▶ For $1 \leq i \leq \ell$, let e'_i be an edge in E such that $\phi(e'_i) = \{v'_{i-1}, v'_i\}$.
- ▶ Since v, v' are adjacent, there exists some edge e'' in E such that $\phi(e'') = \{v, v'\}$.
- ▶ **Idea:** Consider the edge-induced subgraph G'' of G induced by $\{e_1, e_2, \dots, e_k, e'', e'_\ell, e'_{\ell-1}, \dots, e'_1\}$.



Bipartite graphs and odd cycles (continued)

Progress so far: G'' is the edge-induced subgraph of G induced by $\{e_1, e_2, \dots, e_k, e'', e'_\ell, e'_{\ell-1}, \dots, e'_1\}$.

- ▶ **Note:** The sequence of edges $e_1, e_2, \dots, e_k, e'', e'_\ell, e'_{\ell-1}, \dots, e'_1$ must contain an odd cycle as a subgraph!
 - ▶ **Question:** Can you explain why?
 - ▶ Remember: k and ℓ have the same parity.
- ▶ However, we supposed that no subgraph of G is an odd cycle, which is a contradiction.
 - ▶ Hence, any two adjacent vertices must have different colors!
 - ▶ Therefore, our color assignment is a vertex 2-coloring.
 - ▶ This completes the proof for [Part 2]. □

Summary

- ▶ Subgraphs, induced subgraphs
- ▶ Graph colorings, chromatic number, edge chromatic number
- ▶ Brooks' theorem, Vizing's theorem, four color theorem
- ▶ Properties of bipartite graphs